

Children in schoolyards: Tracking movement patterns and physical activity in schoolyards using global positioning system and heart rate monitoring

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ARTICLE INFO

Article history:

Received 29 March 2008

Received in revised form 8 June 2009

Accepted 2 July 2009

Available online 7 August 2009

Keywords:

Children

Schoolyards

Physical activity

Heart rate monitoring

GPS

GIS

ABSTRACT

The aim of this study was to investigate how 6-year-old school children used their schoolyard during recess and how the yard invited physical activity. Correlates for physical activity were in focus for the investigation. Two different schoolyards, one in the city and one rural, in Southern Norway were chosen for the study. The schoolyards were mapped and the affordances for physical activity described and presented in an orthophoto. The children's movement patterns were recorded with portable global positioning system (GPS) and the level of physical activity monitored by heart rate (HR) applying a Polar pulse watch. GPS and heart rate data were transferred to geographical information systems for further analyses and visualisation. The results indicated physical activity (mean heart rate) to be similar for the children in the two schoolyards. Asphalt areas invited running and soccer play and favoured more physical activity in boys than in girls. The rural schoolyard afforded playing in a little forest, which was more attractive to girls than boys. In both schoolyards the children were physically active in free play at moderate to vigorous levels half the time of a recess period of 40 min. The methods of GPS tracking and HR monitoring seemed to be applicable for this purpose. For further studies with young children the equipment needs adjustments. Experiences from this study recommend more attractive landscapes and facilities that afford increased physical activity in schoolyards.

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1. Introduction

Being outdoors is the strongest correlate of physical activity in preschool children (Baranowski et al., 2000; Faber Taylor et al., 2001; Fjørtoft, 2000, 2004). Given the opportunity, children have a natural drive to be active and play freely. Most of children's free play includes a high amount of physical activity and consequently children in free play will be moving. Physical activity is the number one recommendation for a healthy lifestyle throughout the lifespan and giving children healthy habits in early years may give a positive pay off in adulthood (Strong et al., 2005). Several scenarios of children's health reveal a strong tendency of children becoming more inactive, gain weight and being more exposed to lifestyle illnesses such as diabetes 2, coronary diseases, cancer and osteoporosis. Risk of overweight and sedentary behaviour are increasingly evident in children, which has implications for subsequent ages (Ogden et al., 2002; Reilly et al., 2004). Physical activity has proven to be associated with healthy lifestyle and preventing lifestyle diseases (Baranowski et al., 1993; Frost Andersen et al., 2005; Fulton et al., 2001; Strong et al., 2005). Andersen et al. (2006) have described

how moderate to vigorous physical activity is supposed to be effective on children's health and metabolic syndromes. Based on an extensive review of scientific articles, Strong et al. (2005) recommended a minimum of 60 min moderate to vigorous physical activity every day for school-aged youth. This is consistent with the Global Strategy for diet, physical activity and health approved by WHO (2004) and also the Norwegian White Paper on physical activity and health (Action plan for physical activity, 2004).

Australian studies (Salmon et al., 2005) have described how interventions with increased activity in children's leisure time promoted activity and reduced inactivity and TV watching. It is generally accepted that diversity increases activity. The better equipped schoolyards and playgrounds offer a variety of play forms that challenge physical activity. Lindholm (1995) has documented how activities in schoolyards increased with the presence of green structures. Stratton (2000) reports that simple initiatives such as marking the schoolyard with colours have a positive effect on children's physical activity. Zask et al. (2001) found potentials (such as marking hopscotch areas or ball game fields) for increasing physical activity in the schoolyard, particularly for self-organized activities. Simple playground markings and physical structures proved to increase physical activity levels in school children, with a special effect on the inactive children (Ridgers et al., 2007). Some Scandinavian studies have described how diversity of natural

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Table 1

Schools, space and sample of children in the study.

Schools	Area m ²	No. children	Area per child m ²	No. 1st graders	Area per 1st graders m ²	No. samples	No. boys	No. girls
School A	3,200	385	8.3	71	45 (91) ^a	28	10	18
School B	13,000	450	29.0	71	183 (370) ^a	42	21	21

^a Area indicates space per child when only first graders are outdoors, in classes and in smaller groups (in brackets).

environments enhance physical activity and promote motor development in young children (Fjørtoft and Sageie, 2000; Fjørtoft, 2004; Grahn et al., 1997). Those studies showed that natural environments afforded more physical activity than traditional playgrounds. However, determinants for physical activity in children are far from explored and need more investigation. The quantity and quality of outdoor neighborhood environments seem to encourage physical activity in adult people, and contribute to health in late life (Sugiyama and Thompson, 2007). Studies on child–environmental relationships indicated positive influences on children in general (Fjørtoft, 2000; Grahn et al., 1997; Lindholm, 1995) and also on children with attention deficit disorders (Faber Taylor et al., 2001).

2. Issues of research

Behaviour settings are landscapes where activities occur and comprise both the physical and social context of behaviour (Sallis et al., 1998). Such settings are potentially important to individual involvement in physical activity (King et al., 2002). As nearly all children attend school and because they spend a considerable part of their awake time in school, it seems important to understand the impact of the school environment on children's physical activity (Sallis et al., 2001).

In this study we wanted to investigate the affordances of physical activity in two different schoolyards and examine whether different environmental determinants in the schoolyard settings promoted different levels of physical activity in 6-year-old schoolchildren.

2.1. Research questions

The main objective for this study was to describe movement patterns and physical activity levels in 6-year-old schoolchildren in two different schoolyards in Southern Norway. The study also served as a pilot for exploring methods for monitoring such activities through geographical positioning system (GPS) and heart rate monitoring (HR). The objective will be addressed through the following specific research questions:

- How do schoolyards afford physical activity?
- How do first-graders use the schoolyard?
- Which elements or areas determine physical activity?
- How active are 6-year-old in the schoolyard during recess?
- How does free play during recess in schoolyards meet the request of 60 min moderate to vigorous physical activity?
- Are there gender differences in physical activity and intensity in the schoolyards?
- How applicable are monitoring instruments such as GPS and heart rate monitoring for describing children's movement patterns and physical activity?

3. Materials and methods

The methodological approach was a comparative study of children's activities in two schoolyards with different outdoor environments. The affordance of the schoolyards was described by tracking children's movement patterns as well as monitoring their physical activity in the schoolyards. This constituted the basis for data capture.

3.1. Schoolyards and research groups

Two schools with quite different schoolyards were selected and accepted to participate in the study; one city school (A) with a limited asphalt schoolyard, and one rural school (B) with schoolyard extending into a little forest. The two schools were part of the same municipality, but represented quite different locations and designs. School A was a typical old city school and school B was a more modern school located in a rural area. Both schools were primary schools comprising grades 1–7. The first graders, in total 71 six-year-old at each school, were selected as research groups. Consent was asked from the parents and totally 70 children who got the consent participated in the study. At school A 28 children, 10 girls and 18 boys, constituted the sample. At school B 42 children, 21 girls and 21 boys participated (Table 1). Tracking of children's movements and monitoring physical activity were performed during a 4 weeks period in September in good weather. The children were monitored in smaller groups of maximum 12 children at a time due to limited number of monitoring equipments. Monitoring was conducted during recess periods of 40 min at both schools and each child was monitored only once. All monitoring and data collection and analyses were treated anonymously due to directions from the Norwegian Data Inspectorate.

3.2. Defining determinants for physical activity

Certain areas of the schoolyards were classified and screen digitized based on ortophotos. Both schoolyards were divided into two main areas; schoolyard A into a small “soccer field” and the remaining asphalt area (Fig. 1), schoolyard B into a forest and an asphalt area (Fig. 2). Spatial observations located to those areas were selected for further statistical analyses. A set of other interesting areas were also classified, such as sandpits, swing, seesaw, etc., but later found to be too small to be used in spatial analyses, due to limited accuracy of the GPS units used.

3.3. Measuring movement patterns and physical activity

Children's movement patterns and physical activity were measured during recess periods of 40 min in the middle of the day at both schools. Global positioning system (GPS) was applied for tracking children's movement patterns in the schoolyards. Portable GPS units (Garmin Forerunner, 2006) were fixed to the wrist of the children, recording geographical positions from satellites every 5th second. The GPS units measured positions through four or more satellites. Problems occurred when the children moved close to tall buildings especially in a north position, giving abnormal outliers. Default observations outside the marked areas in Figs. 1 and 2 were removed from the data set. In school A, 1163 out of 10791 observations (10.7%) and in school B, 1093 out of 17802 (6.1%) observations were removed.

Garmin Forerunner 201 Owner's Manual (2006) states GPS accuracy to be less than 15 m root mean square (RMS) 95% of measured time. There are good reasons to believe that this is a conservative estimate. A recent study, using the same GPS units, found that for static measurements the interval correct value ± 4 m contains 95% of all measurements (Sageie et al., 2007). The inaccuracy caused problems in monitoring activity at small playground areas such as

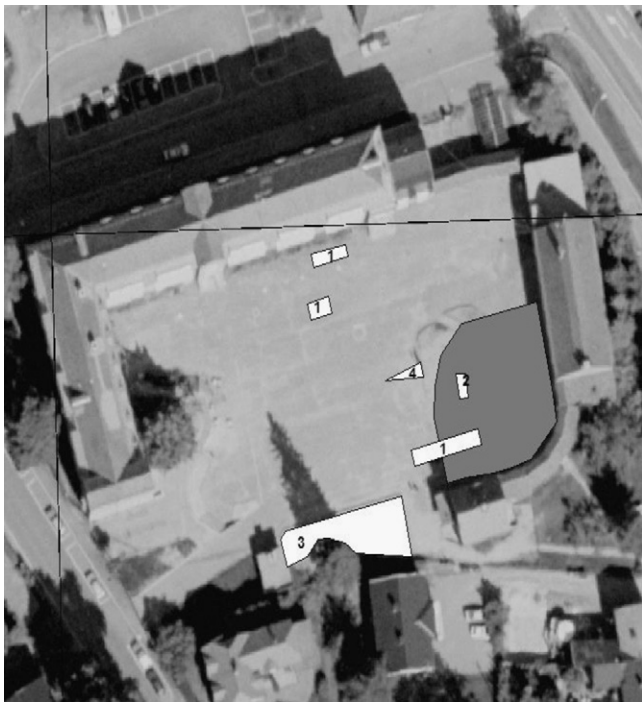


Fig. 1. Schoolyard A, the city schoolyard with classified activity areas: 1: hopscotch, 2: swing, 3: sandpit and play equipments, 4: basketball goal. Grey coloured area: area for soccer kicking at goals.

the minor play equipments and small marked spots (Fig. 1). With these reservations, the GPS units used in this study were considered to be applicable for scientific use (Sageie et al., 2007).

Physical activity was recorded as heart rate data monitored by simple chest electrodes, where signals were transferred and stored in a Polar wrist watch (Polar 521). Heart rate was monitored every 5th second and stored in the pulse watch (Polar Electro Oy, Kempele, Finland). Nine cases (3 from school A and 6 from school B) with incomplete heart rate and/or positional measurements were excluded. The resulting 61 cases were the basis for statistical analyses.

3.4. Data capture and statistical analyses

Heart rate data and coordinates from GPS tracking were temporarily stored in a solid state memory and later transferred to a study database (Garmin and Polar software) and for further analyses in statistical packages. GPS data and heart rate data were also transferred to Geographical information systems (GIS) software (ArcView GIS 3.3, ESRI) (Lo and Yeung, 2006; Worboys and Duckham, 2004).

Physical activity was measured as heart rate (HR). Maximum heart rate (MHR) was estimated to be 220 beats per minute (bpm), as values of MHR in children are often measured up to 220 bpm (Åstrand et al., 2003). The average heart rate for each child during the recess period was calculated. Average heart rate values were used in statistical analyses.

Four intensity levels were used as threshold values to represent low physical activity (LPA < 120 bpm), low to moderate physical activity (LMPA = 120–140 bpm), moderate to vigorous physical activity (MVPA = 140–160 bpm) and vigorous physical activity (VPA > 160 bpm), respectively (adjusted after Stratton, 2000; Ridgers et al., 2007). Total time and percentage of time each child spent at LPA, LMPA, MVPA and VPA during recess were calculated and used in subsequent analyses.

By combining GPS data of movement patterns with HR data in thematic maps children's movement patterns were explored, and the correlations between landscape elements, spatial locations and physical activity were identified. Total time and percentage of time each child spent at different areas of the schoolyards were calculated.

Time spent at each intensity level and in each area were calculated in percentages, since the monitoring period for each child was not identical (mean = 38.6 min, SD = 7.8).

Statistical analyses were performed by using Minitab and the Statistical Packages for the Social Sciences, SPSS/PC+: frequency analyses, means, normality tests, *t*-tests, Mann–Whitney (Wilcoxon), ANOVA General Linear Model.

4. Results

4.1. The schoolyards

Sample school A was 100 years old and located in the city, comprising 385 children. A sample of 28 first graders, 10 boys and 18 girls out of totally 71, participated in the study. The schoolyard was mostly flat and covered with asphalt. One little artificial hill was constructed in one corner and some play equipment for the youngest ones (swing, seesaw, climbing nets and a sandpit) located in another corner. The surface of the flat ground was covered with asphalt as shown in Fig. 1. The total area was 3200 m², which

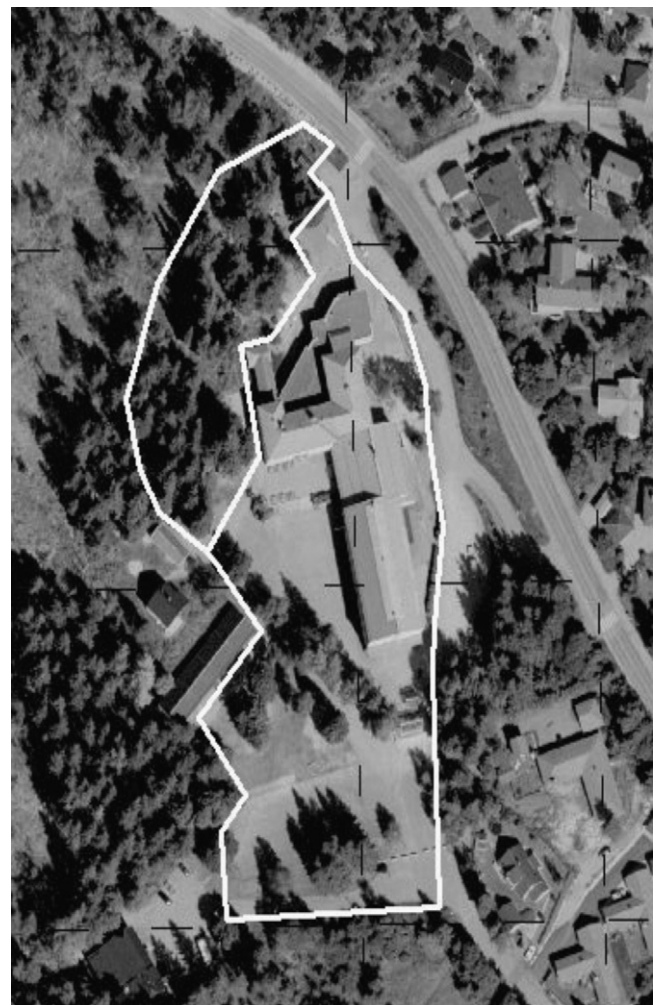


Fig. 2. Schoolyard B, the rural schoolyard. Lined areas show the area of the forest and the asphalt yard.

afforded 8.3 m² of space for each child when all children were out at the same time. When only the first graders were out in recess the space was 45 m² per child. When being outdoors in smaller groups, each child managed a space of 91 m² (Table 1).

School B was located at the rural side of the municipality, where the schoolyard comprised a total area of 13,000 m² including a little forest (see Fig. 2). The traditional part of the schoolyard had the facilities of a soccer field, and play equipment for the younger children, while flat areas were covered with asphalt. In total there were 450 children at school B, which allowed a space of 29 m² per child when all the children were out in recess. A sample of 42 first graders, 21 boys and 21 girls out of totally 71, participated in the study. When all the first graders were out in recess the space was 183 m² per child. When being outdoors in smaller groups, each child managed a space of 370 m² (Table 1).

Both schoolyards were equipped with playground apparatuses for the 6-year-old, which afforded swinging, seasawing, climbing and playing with sand. Asphalt areas afforded running, soccer and other ballgames (Figs. 1 and 2). Schoolyard B had a soccer field in the end of the yard (located in the lower part of Fig. 2), but the first graders had little tradition in using this part of the schoolyard. Schoolyard B afforded a forest landscape in addition to traditional playground equipment. The little forest was frequently used by the children as it was dedicated to the first graders as their preferred playground area. School B benefited also from larger space and more diversity in landscape elements (Fig. 2).

Schoolyard A typically afforded traditional playground activities determined by the available play equipment and an open asphalt field with no markings or structures that invited running activities.

4.2. Movement patterns in the two schoolyards

The environments of the two schoolyards represented different landscapes which typically afforded different opportunities of activity. The flat asphalt area in schoolyard A invited running around and casual use of play equipment: sandpit, swing, seesaw, and the climbing apparatus. Figs. 3 and 4 show typical movement patterns of a girl and a boy in schoolyard A.

Boys tend to roam all over the schoolyard while the girls are more moving across, visiting the area of the swing, the sandpit and the play equipments (Fig. 1). Some eager soccer players occupied fences and corners for kicking at goals. Such activities provoked high heart rates, exemplified in Fig. 5 for a boy having a mean HR of 173 bpm during a recess period of 40 min.

In schoolyard B there were two main areas used by the 6-year-old, the forest and the asphalt areas around the school as shown in Fig. 2. Typical movement patterns for boys and girls are illustrated in Figs. 6 and 7 indicating that girls favoured the forest area more than boys, while the boys used the asphalt areas around the school building more.

4.3. Physical activity in the two schoolyard environments

Table 2 shows mean heart rate for boys and girls for school A and B, obtained by first calculating the average heart rate for each child during the recess period (12 heart rate observations per minute for approximately 40 min), and then taking the mean of the average heart rates.

Physical activity levels seemed to be quite similar in the 6-year-old in the two schoolyards. Mean heart rate was 146.2 bpm (SD 17.0) for school A and 147.2 bpm (SD 14.3) for school B. The differences were not significant ($p < 0.05$, two samples t -test).

In both schoolyards the levels of physical activity differed between boys and girls, boys being more physically active than girls in schoolyard A, while the girls were more physically active than

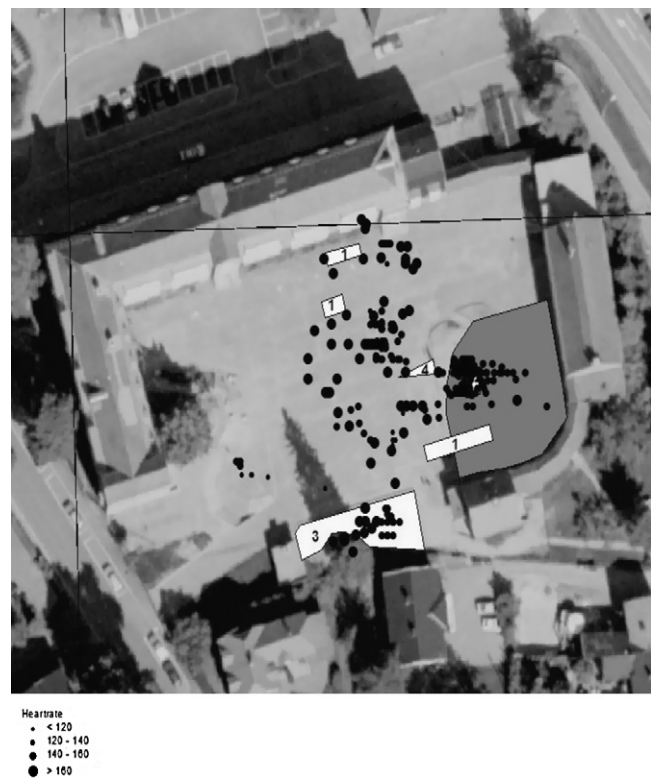


Fig. 3. Typical movement pattern of a 6-year-old girl in schoolyard A monitored by GPS. The size of the bulbs indicate intensity levels of activity, monitored as heart rate.

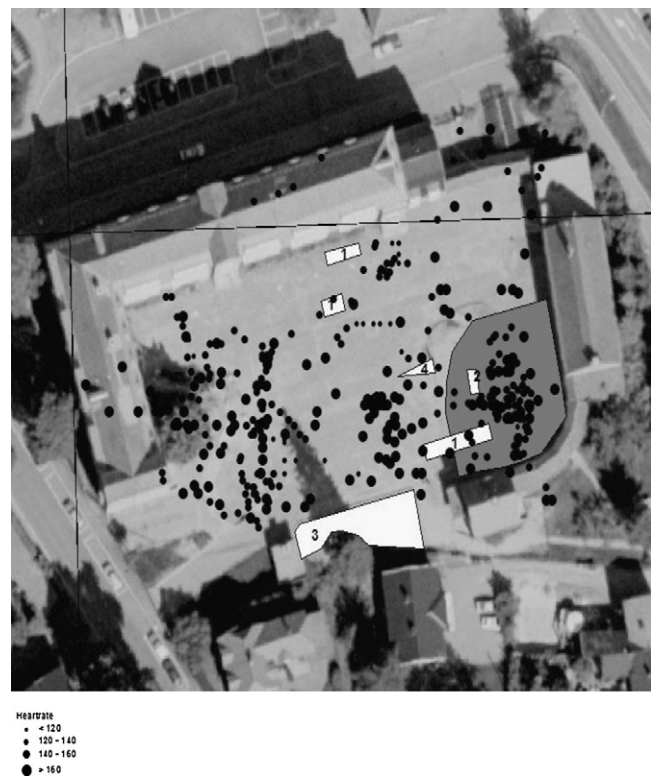


Fig. 4. Typical movement pattern of a 6-year-old boy in schoolyard A monitored by GPS. The size of the bulbs indicate intensity levels of activity, monitored as heart rate.

Table 2

Mean heart rate in 6-year-old children during recess at sample school A and B.

Schools	N	Mean	Minimum	Maximum	Std. Deviation	Std. Error mean
A						
Total sample	25	146.22	114.64	177.41	17.04	3.41
Boys	9	154.80	122.71	177.41	19.58	6.53
Girls	16	141.40	114.64	166.05	13.83	3.46
B						
Total sample	36	147.18	103.43	174.68	14.27	2.38
Boys	19	146.21	103.43	174.68	15.91	3.65
Girls	17	148.27	132.37	168.59	12.58	3.05

boys in schoolyard B. The differences were however not significant ($p < 0.05$, two samples t -test).

Comparing physical activity between boys in the two schoolyards, boys in the squared asphalt schoolyard A had a higher level of physical activity than boys in schoolyard B. The physical activity levels for girls were opposite, as the girls in the forest yard B were more physically active than girls in schoolyard A. But again the differences were not significant ($p < 0.05$, two samples t -test).

The possible interactions between school and gender were analysed with ANOVA General Linear Model. The result showed a weak but not significant interaction between the groups ($p = 0.06$, $F = 3.67$, d.f. = 1).

4.4. How active are the children?

Fig. 8 shows the mean percentage of time the 6-year-old spent at different intensity levels of physical activity in the two schoolyards, obtained by first calculating the percentage of time each child spent at each intensity level. The children in the two schoolyards were physically active in self-organized activities at moderate and vigorous levels in approximately half the time of a recess period of 40 min.

A significant difference ($p < 0.05$, two samples t -test) between boys and girls at VPA level at school A was observed (Fig. 9). The boys spent more time in vigorous physical activity (46.2%) than the girls (23.4%). Adding up the two activity levels of MVPA and VPA the boys were at these levels in 71.6% of the time, while the girls spent 53.1% of the time at these intensity levels.

At school B boys and girls were more equal in time spent in MVPA and VPA. They spent 56.4% (boys) and 61.6% (girls) of the time at these intensity levels (Fig. 10).

There were also observed differences between girls at the two schools in the lowest activity level (Figs. 9 and 10), showing that the girls in schoolyard A were more in the LPA level (21.3%) than girls at schoolyard B (10.4%). These differences were however not significant ($p < 0.05$, Mann–Whitney).

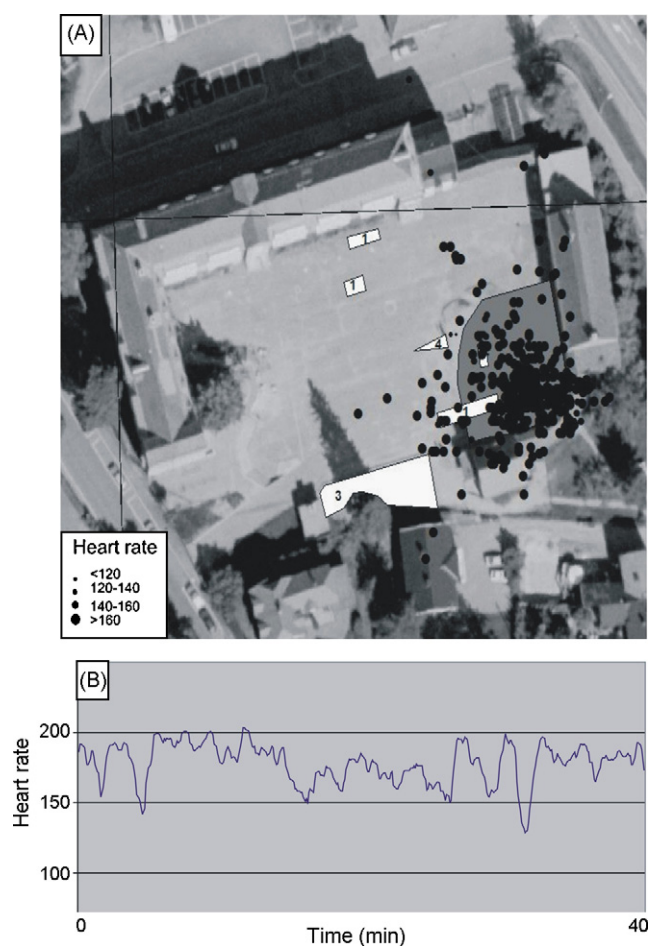


Fig. 5. The movement pattern and intensity levels of a 6-year-old boy kicking at goals in schoolyard A during recess. (A) The size of the bulbs indicate intensity levels of activity, monitored as heart rate. (B) The heart rate curve showing variations of HR 150–210 bpm during 40 min of physical activity, mean HR = 173 bpm.

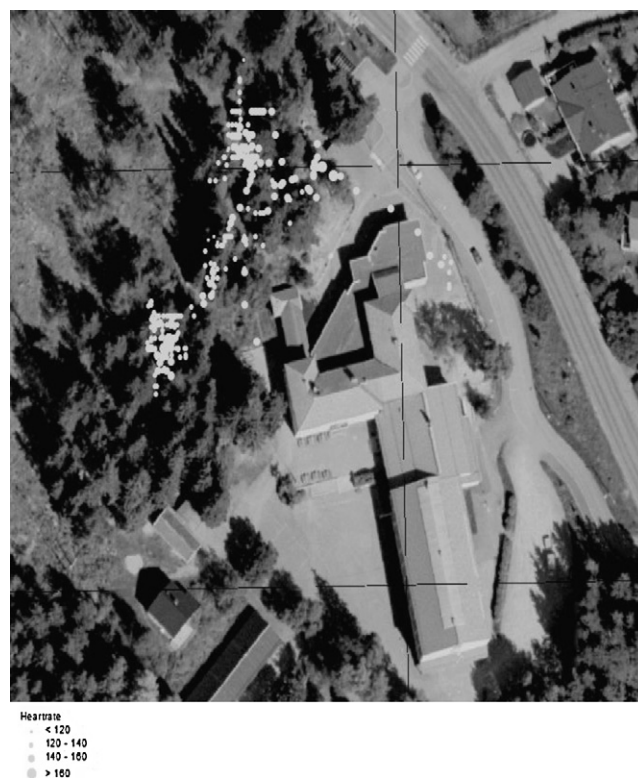


Fig. 6. Typical movement pattern of 6-year-old girl in schoolyard B monitored by GPS. The size of bulbs indicate intensity levels of activity, monitored as heart rate.



Fig. 7. Typical movement pattern of 6-year-old boy in schoolyard B monitored by GPS. The size of bulbs indicate intensity levels of activity, monitored as heart rate.

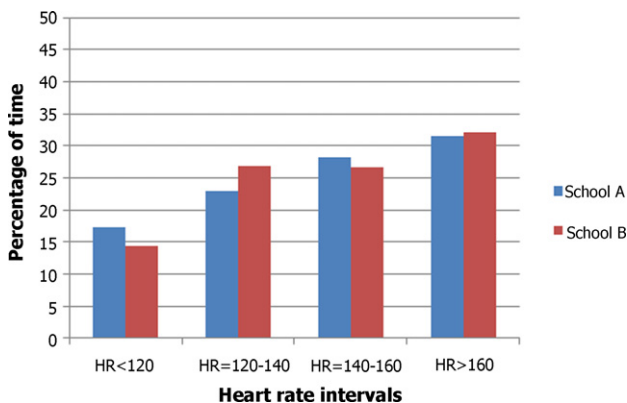


Fig. 8. Distribution of time spent (percentages) at different intensity levels in 6-year-old children, expressed as heart rate intervals at the sample schools.

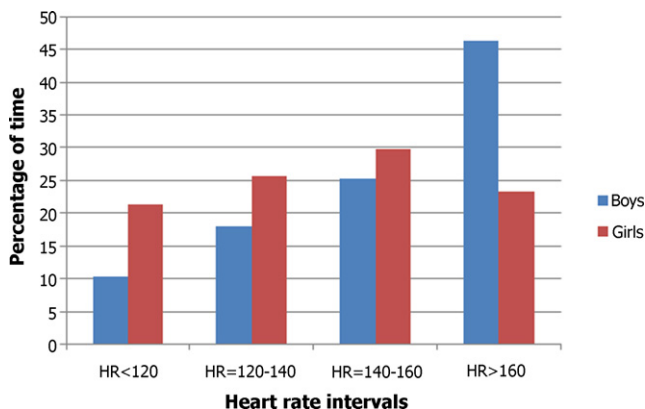


Fig. 9. Time spent (percentages) at different intensity levels, expressed by heart rate intervals of boys and girls in school A.

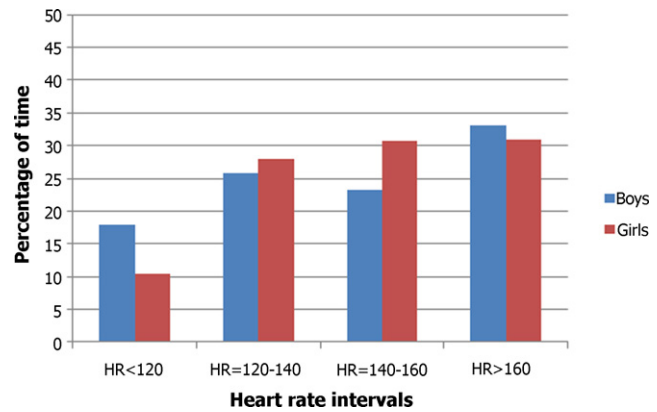


Fig. 10. Time spent (percentages) at different intensity levels, expressed by heart rate intervals of boys and girls in school B.

4.5. Determinant areas for physical activity

The schoolyards were classified into activity areas. Schoolyard A was classified into an asphalt area and a soccer area (Fig. 1), while schoolyard B was classified into a forest and an asphalt area (Fig. 2). Table 3 shows the mean percentage of time the children spent in different parts of the schoolyards, obtained by first calculating the percentage of time each child spent in these areas.

In school A, the boys spent in average 20.0% of their time in the soccer area and 80.0% of their time in the rest of the schoolyard, whereas the girls only spent 9.9% of their time in the soccer area. Not surprisingly, boys seem to use the soccer area more than girls. Note that the asphalt area is much larger than the soccer area, which explains the relatively low percentages of time spent in the soccer area both for boys and girls.

In schoolyard B the girls spent a larger percentage of their time in the forest area (51.7%) than the boys (32.1%). These differences were significant ($p < 0.05$, Mann-Whitney).

5. Discussion

5.1. The affordance of schoolyards

The two schoolyards afforded different types of play and physical activity during recess periods. Schoolyard A was rectangular with a total area of 3200 m², providing the children a space of 8.3 m² when all children in the school have recess at the same time. As the first graders had their own recess period of 40 min each day the space provided was then 45 m² when all the first graders were out at a time and 91 m² when they were outdoors in smaller groups, which was common practice. Therefore, space did not seem to be a limit to outdoor activities. The yard was covered by asphalt and afforded typically the activities a traditional playground may provide. There were few appropriate landscape structures that afforded activity in the flat asphalt schoolyard, and therefore the movement pattern became naturally more like roaming around. The playground equipment such as sandpit, climbing constructions, seesaws and swings did not have a very high attraction. Few indicators or struc-

Table 3

Percentage of time spent by girls and boys in different areas in the two schoolyards.

School	Sex	Asphalt	Forest	Soccer
A	Girls	90.1		9.9
	Boys	80.0		20.0
B	Girls	48.3	51.7	
	Boys	67.9	32.1	

tures clearly invited physical activity except for the “soccer corner” (Fig. 1). A few boys were eager to play soccer at targets located near a fenced corner (an arcade) of the schoolyard. This structure seemed to invite a particular soccer activity “kicking at a goal”, as there were no proper soccer goals in the schoolyard. It was obvious that this arcade became a specific determinant for vigorous activity in some of the boys (Fig. 5). Girls in school A did not have any special physical activities except for using the playground equipment, which seemed not to be very attractive.

Schoolyard B had more diversity in its surroundings as it included a little forest (Fig. 2) that was extensively used during recess. This schoolyard provided large space of totally 13,000 m² which afforded each child a space of 370 m² being outdoors in smaller groups. According to Gibson's theory of affordances (Gibson, 1979) children perceive landscapes and landscape elements as functions for use, and thus diversity in landscape structure promoted activity. Previous studies by Fjørtoft and Sageie (2000) documented how diversity in outdoor environments enhanced physical activity and promoted motor development in small children and Lindholm (1995) described how green structures in and around schoolyards may enhance play activity in children.

Although Schoolyard B was hypothesised to afford more possibilities for physical activity, no significant differences in activity intensity monitored as mean heart rate over a recess period of 40 min were detected between the 6-year-old at the two schools. Possible explanations for such findings may be that physical activity was monitored as intensity (HR) and not diversity in activity and motor skills. As high intensity is linked to VPA and high heart rate levels, those were most often found when the children were moving at high speed. Open space afforded running, as was demonstrated in schoolyard A. Schoolyard B afforded more options for play, but did not result in higher activity, monitored as heart rate.

Small but not significant gender differences were detected. In schoolyard A girls were less active than boys, while in schoolyard B the girls were more active than the boys (Table 2). In schoolyard A boys seemed to be more involved in soccer play, while the yard did not have any specific affordances for the girls such as play hopscotch or skipping rope or other traditional games. Several studies have documented that simple initiatives in schoolyards such as markings and organized activities have increased children's activities during recess (Stratton, 2000; Zask et al., 2001; Ridgers et al., 2007). In schoolyard A there was a lack of such markings and a rehabilitation of the hopscotch areas should be a potential for increasing physical activity especially in the girls. A recent study of Haug et al. (2008) found that children in secondary schools with many facilities had 4.49 times higher odds of being physically active compared to children in schools with fewer facilities. Lindholm (1995) and Fjørtoft and Sageie (2000) have described how green structures may motivate children's play and physical activity. This may explain the higher activity of girls in schoolyard B as they used the forest area more than the boys (Table 3). It is generally accepted that green structures have a considerable potential for promoting physical activity and improving health in adult people (Tzoulas et al., 2007). Previous studies have documented that stimulating green environments also influence children's play and physical activity (Lindholm, 1995; Fjørtoft, 2004).

5.2. Intensity and duration of physical activity in recess periods

The 6-year-old children at the two schools were out in the schoolyard during recess periods of 40 min every day. The activities were self-organized and the facilities of the schoolyards afforded opportunities for activity. Fig. 8 shows the percentage of time children spent at different intensity levels. At both schools the children were active at VPA level more than 30% of their outdoor time and adding up the MVPA and the VPA levels those levels covers a total

of 59.8% of outdoor time at school A and 58.3% at school B. Gender differences in time spent at different activity levels were observed. Significant differences in VPA level were observed between boys and girls in schoolyard A, the boys spending 46.2% of recess time at this intensity level. Summing up the MVPA and VPA levels for schoolyard A the boys were active at these levels 71.6% of total outdoor recess time and the girls were active at these levels 53.1% of the time. The gender differences in schoolyard B were not significant, but here the girls were more active (61.2% of time) than the boys (56.4% of time) at moderate and high intensity levels (Figs. 9 and 10). It should also be noticed that the girls in schoolyard A were observed more in lower intensity levels than girls in schoolyard B, although these differences were not significant.

From a health perspective, recommended levels of physical activity in children are a minimum of 60 min of moderate and high intensity every day (Strong et al., 2005), while Andersen et al. (2006) discuss whether 90 min of moderate or vigorous physical activity should be recommended. The children were in average at these intensity levels for 22 min at school A and 24 min at school B. In this perspective the effective contribution of physical activity during recess in the two schoolyards may be a significant contribution, but far from sufficient. As school is compulsory to all children it could be an option to school authorities to provide extended or multiple recess periods during the school day to provide all children the recommended time for physical activity. To guarantee a MVPA level of activities the schoolyards must be planned and facilitated for more physical activity and considered in an age and gender perspective. This should give planners and landscape architects a novel opportunity to design schoolyards as landscapes that challenge a diversity of movements and physical activity that are adapted to different age groups and gender.

5.3. Determinants for physical activity

Landscape elements that afforded MVPA and VPA in this study proved to be structures inviting locomotion in high speed, whether in running or playing soccer, which were typically attractive to boys more than girls (Table 3). Determinants that challenged MVPA and VPA were identified as open space and target areas for ballgames such as kicking at goals as seen in schoolyard A. The soccer field in schoolyard B was not used by the first graders, because they from habitual routines more often used the forest area for outdoor play. Six-year-old children seem to be physically active wherever there is space for running, and they seem to appreciate such affordances. Forest areas seemed to invite more physical activity in girls than in boys, but green landscapes are desired options for play in both genders (Lindholm, 1995). Whether other determinants such as markings for hopscotch, rope jumping and areas for other ballgames promote physical activity should be more extensively explored as indicated by Haug et al. (2008). Development of a new conceptual framework for understanding determinants for children's physical activity in schoolyards are welcomed, as suggested by van Kamp et al. (2003). Indicators for more vigorous activity are those that promote running, jumping, climbing, kicking and throwing. But also landscape structures or equipment that challenge balance, coordination, force and agility are important elements in landscapes for diverse activities.

5.4. Applicability of GPS and heart rate monitoring instruments

Measuring heart rate is a significant and well known metrics to determine the intensity of physical activity (Åstrand et al., 2003). The monitoring equipment was however not well adjusted to small bodies, as the chest belts in some cases lost contact with the body and signals stopped. When this problem occurred the case was excluded.

Using GPS and GIS to track and visualize the children's positions, movement and area of playing seemed promising. Tall buildings and dense vegetation may disturb the GPS signals and lead to inaccuracy and outliers. As mentioned earlier, positional accuracy represented a limitation for the scale and resolution of the area under investigation. This made it difficult to identify activities at play equipments such as swings, seesaws and climbing houses.

Using two separate equipments to record positions and heart rate was cumbersome. More modern equipments are now available for recording positions and heart rate in one unit and they are more robust concerning positional accuracy.

6. Conclusion and recommendations

The present study documented how 6-year-old children utilised the affordances of two different schoolyards. The schoolyards provided different space and qualities of outdoor environments, but those factors did not seem to affect the physical activity level of the children in general. A gender difference in activities was observed. The flat, asphalted schoolyard favoured boys playing soccer, while the forest yard seemed to favour higher intensity of physical activity in girls. Open space favoured locomotion and moderate to vigorous physical activity in the asphalt schoolyard, while running around and exploring the forest area were documented by movement patterns that enhanced more moderate levels of physical activity.

Methods of GPS for mapping movement patterns in the two schoolyards were combined with monitoring heart rate by a Polar Pulse watch. This enabled tracking of children's movement in the schoolyards and location of areas that promoted different activity levels. This method is not widely explored, and more accurate equipment is now being developed, which seems promising for future studies.

More research is needed to identify environmental indicators for physical activity in children at different gender and age groups. Hopefully, this study will inspire researchers, planners and designers in developing schoolyards into appropriate and challenging landscapes that promote more and diverse physical activity in school-aged children.

Acknowledgements

The authors want to acknowledge children, teachers, school administrators as well as the planning and survey department in the municipality for positive and active cooperation in this project. We also want to thank Per Chr. Hagen for helpful support on statistical analysis.

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